Phase 1 — Reference P&ID Analysis (AD_01.10 / 11 / 14 / 17)

Extracted best-practice conventions from the SCK CEN / Mott MacDonald example drawings to drive the v4 refinement.

Source drawings reviewed

Drawing	Content	Size
AD_01.10 P&ID PCW (TP2=PS01.PAB12)	Process Cooling Water P&ID (2 sheets: notes + diagram)	A4/A3
AD_01.11 P&ID Instrument Air (TP2=PS01.QFB10)	Instrument-air P&ID (3 sheets)	A4/A3
AD_01.14 Piping Layout - Instrument Air (PS05.QFB10)	Piping GA / layout	A3
AD_01.17 Instrument Air - Piping Routing & Layout	Routing layout	A0

1. Terminal-point (TP) placement — KEY FINDING

Terminal points sit **at the left and right page edges** (and occasionally top/bottom for vertical flows). Each TP is a small assembly, not a bare diamond:

```
     ┌────────────────────────────┐  ← scalloped "cloud" callout
     │         WATER GLYCOL        │     · system / medium name (top line)
     │  ◁ =NP.PS01_PFB503 │ 7884 │     · arrow box: destination DWG ref + line no.
     │   FROM MAC (WEST GALLERY) │     · TO/FROM <area> annotation (bottom line)
     └────────────────────────────┘

               ◇ TP.PS01.4002         ← TP diamond w/ TP code + category flag (△B) +
NOTE ref
                │
        ────────┘ (pipe continues into / out of the drawing)
```

- **Arrow direction encodes flow**: left-pointing arrow = incoming (`FROM`), right-pointing = outgoing (`TO`).
- The pipe **runs to the frame border** with a short connection stub.
- “FROM SHEET 2 / TO SHEET 2” clouds appear at the **bottom edge** for inter-sheet continuation.
- Each TP carries a **category triangle** (△B etc.) and a **NOTE n** reference.

2. Line naming strategy

Lines are labelled **inline, sitting on the pipe** in a small white gap:
`BR0021-4"-PAC1-BA` = line-number – size – fluid/spec-class – insulation/route code.
Labels repeat along long runs and at every branch. This is the mechanism the reference uses to keep a **monochrome** drawing readable — the **name on the line**, not colour, is the primary identifier.

3. White boxes for tags

Every instrument / valve tag (e.g. `AA0019`, `AA3001 / MOV-03`) sits in a small **white rectangle** so it never collides with piping behind it. Tag boxes are opaque white with a hairline border and live on the topmost layer.

4. Valve orientation

- Valves are drawn **in-line** with the pipe (gate-valve bow-tie), white-filled with a black outline so they read clearly where they cross a coloured/black line.
- Actuated valves (`MOV`, control) add the actuator glyph above the body.
- Tags are placed in white boxes directly **below** the valve.

5. Other conventions adopted

- Scalloped “cloud” callouts for off-sheet/area references.
- Category triangles (△B/C/E/…) used as note + scope flags.
- Reference-drawings list and a “Key to symbols” cell in the title strip.
- Branch take-offs carry their own line number.

How v4 applies these

Reference convention	v4 implementation
TP at L/R edges with cloud + diamond + arrow	`terminal_point_edge()` on layer `02_TerminalPoints_EDGE`; stacked along frame edges
Inline line names (mono legibility)	new layer `04C_Piping_LINENAMES` + `line_label()`
White tag boxes	`tag_with_box()`; instrument/valve tags on `12_Tags_Instruments` (front)
In-line + control-view valves	primary in-line valves + `08B_Valves_HORIZONTAL_OVERLAY` tracked-asset row
Flow-direction arrows	arrow heads in TP assemblies (TO/FROM)